

Couse Code: NECE102



Dr. Swati Rajput

Assistant Professor

Department of Electronics Engineering

Indian Institute of Technology (Indian School of Mines) Dhanbad

Digital Electronics: Course Goals and Key Takeaways



Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance



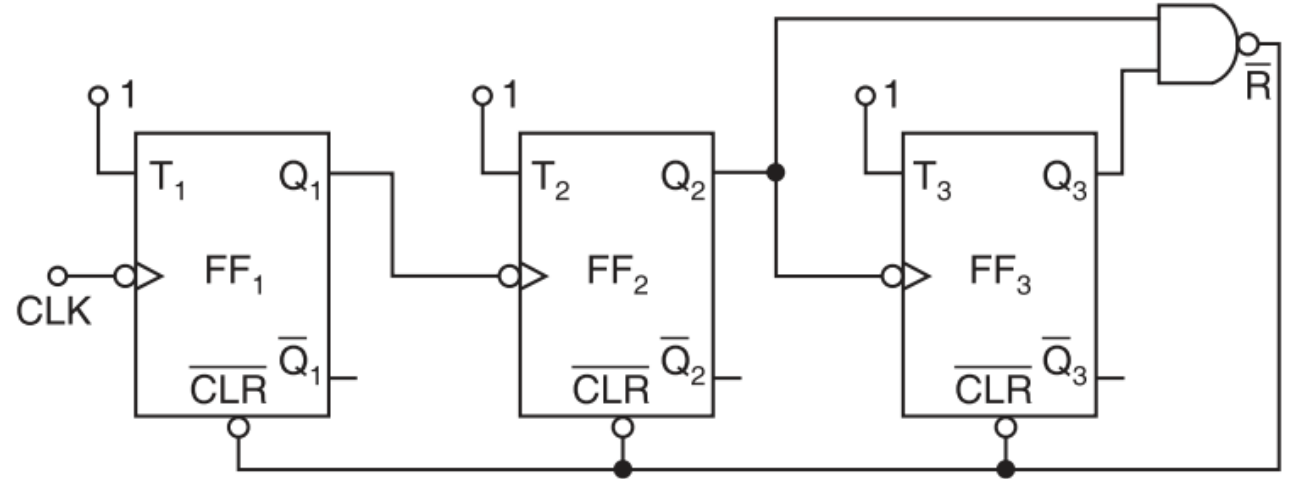
Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:	Reference Books:
1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.	1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015. 2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Mod-6 Asynchronous Counter using Flip Flops



After pulses	State			R
	Q_3	Q_2	Q_1	
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
↓ ↓ ↓				
7	0	0	0	0
	0	0	1	0

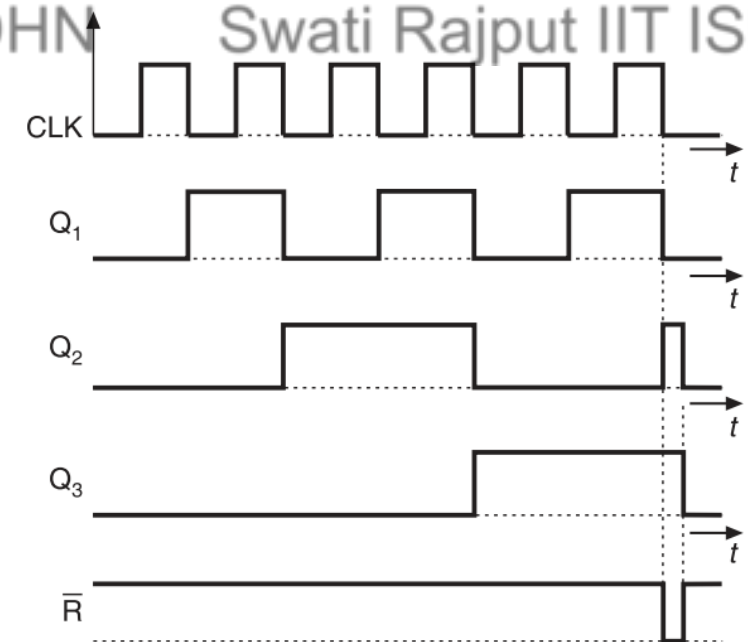


- ❖ With the present state outputs Q_3 , Q_2 and Q_1 as the variables, and reset R as the output and obtain an expression for R in terms of Q_3 , Q_2 and Q_1 . That decides the feedback to be provided.
- ❖ From the truth table, $R = Q_3 Q_2$. For active-LOW reset, $\bar{R}$ is used.
- ❖ The reset pulse is of very short duration, of the order of nanoseconds and it is equal to the propagation delay time of the NAND gate used. The expression for R can also be determined as follows.

$R = 0$ for 000 to 101, $R = 1$ for 110, and $R = X$ for 111

Therefore,

$$R = Q_3 Q_2 \bar{Q}_1 + Q_3 Q_2 Q_1 = Q_3 Q_2$$



(b) Timing diagram

Mod-10 Asynchronous Counter using Flip Flops

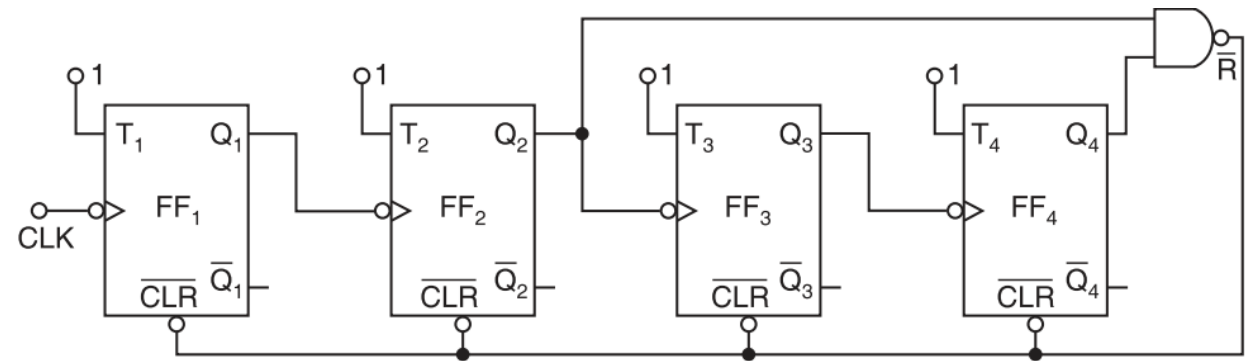
- ❖ A mod-10 counter is a decade counter. It is also called a BCD counter or a divide-by-10 counter.
- ❖ It requires four FFs (the smallest value of n satisfying the condition $10 \leq 2^n$ is $n = 4$). So, there are 16 possible states, out of which ten are valid and the remaining six are invalid.
- ❖ The counter has ten stable states, 0000 through 1001, i.e. it counts from 0 to 9.
- ❖ The initial state is 0000 and after nine clock pulses it goes to 1001.
- ❖ When the tenth clock pulse is applied, the counter goes to state 1010 temporarily, but because of the feedback provided, it resets to initial state 0000.
- ❖ So, there will be a glitch in the waveform of Q_2 . The state 1010 is a temporary state for which the reset signal $R = 1$, $R = 0$ for 0000 to 1001, and $R = X$ (don't care) for 1011 to 1111.
- ❖ From the K-map, $R = Q_4Q_2$. So, feedback is provided from second and fourth FFs.

After pulses	Count			
	Q_4	Q_3	Q_2	Q_1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

(a) Count table

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

(b) K-Map



(c) Logic diagram



- ❖ Asynchronous counters are serial counters. They are slow because each FF can change state only if all the preceding FFs have changed their state.
- ❖ The propagation delay thus gets accumulated, and so causes problems. If the clock frequency is very high, the asynchronous counter may skip some of the states and, therefore, malfunction.
- ❖ This problem is overcome in synchronous or parallel counters. **Synchronous counters** are counters in which all the FFs are triggered simultaneously (in parallel) by the clock-input pulses.
- ❖ Whether an FF toggles or not depends on the FF's inputs (J, K, or D, or T, or S, R).
- ❖ Since all the FFs change state simultaneously in synchronization with the clock pulse, the propagation delays of FFs do not add together (as in ripple counters) to produce the overall delay.
- ❖ In fact, the propagation delay of a synchronous counter is equal to the propagation delay of just one FF plus the propagation delay of the gates involved. So, the synchronous counters can operate at much higher frequencies than those that can be used in asynchronous counters.
- ❖ Synchronous counters have the advantages of high speed and less severe decoding problems, but the disadvantage of having more circuitry than that of asynchronous counters.

Design of Synchronous Counter



For a systematic design of synchronous counters, the following procedure is used.

Step 1. Number of flip-flops: Based on the description of the problem, determine the required number n of the FFs—the smallest value of n is such that the number of states $N \leq 2^n$ —and the desired counting sequence.

Step 2. State diagram: Draw the state diagram showing all the possible states. A state diagram, which can also be called the transition diagram, is a graphical means of depicting the sequence of states through which the counter progresses. In case the counter goes to a particular state from the invalid states on the next clock pulse, the same can also be included in the state diagram.

Step 3. Choice of flip-flops and excitation table: Select the type of flip-flops to be used and write the excitation table. An excitation table is a table that lists the present state (PS), the next state (NS) and the required excitations.

Step 4. Minimal expressions for excitations: Obtain the minimal expressions for the excitations of the FFs using the K-maps drawn for the excitations of the flip-flops in terms of the present states and inputs.

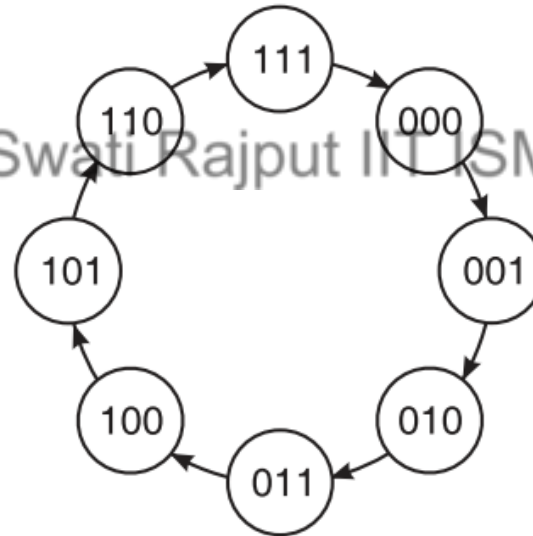
Step 5. Logic diagram: Draw a logic diagram based on the minimal expressions.

Design of Synchronous 3- Bit Up Counter



Step 1. Determine the number of flip-flops required: A 3-bit up-counter requires 3 flip-flops. The counting sequence is 000, 001, 010, 011, 100, 101, 110, 111, 000 ...

Step 2. Draw the state diagram: The state diagram of the 3-bit up-counter is drawn as shown below



(a) State diagram

Design of Synchronous 3- Bit Up Counter



Step 3. Select the type of flip-flops and draw the excitation table: JK flip-flops are selected and the excitation table of a 3-bit up-counter using J-K flip-flops is drawn as shown

PS			NS			Required excitations					
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	1	0	x	0	x	1	x
0	0	1	0	1	0	0	x	1	x	x	1
0	1	0	0	1	1	0	x	x	0	1	x
0	1	1	1	0	0	1	x	x	1	x	1
1	0	0	1	0	1	x	0	0	x	1	x
1	0	1	1	1	0	x	0	1	x	x	1
1	1	0	1	1	1	x	0	x	0	1	x
1	1	1	0	0	0	x	1	x	1	x	1

(b) Excitation table

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

PS			NS			Required excitations					
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	1	0	x	0	x	1	x
0	0	1	0	1	0	0	x	1	x	x	1
0	1	0	0	1	1	0	x	x	0	1	x
0	1	1	1	0	0	1	x	x	1	x	1
1	0	0	1	0	1	x	0	0	x	1	x
1	0	1	1	1	0	x	0	1	x	x	1
1	1	0	1	1	1	x	0	x	0	1	x
1	1	1	0	0	0	x	1	x	1	x	1

(b) Excitation table

Design of Synchronous 3- Bit Up Counter



Step 4. Obtain the minimal expressions: From the excitation table it is seen that, $J_1 = K_1 = 1$, because all the entries for J_1 and K_1 are either a 1 or an X. The K-maps for excitations based on the excitation table and the minimal expressions for excitations J_3 , K_3 , J_2 , and K_2 in terms of the present outputs Q_3 , Q_2 , and Q_1 obtained by minimizing the K-maps are shown below.

PS			NS			Required excitations					
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	1	0	x	0	x	1	x
0	0	1	0	1	0	0	x	1	x	x	1
0	1	0	0	1	1	0	x	x	0	1	x
0	1	1	1	0	0	1	x	x	1	x	1
1	0	0	1	0	1	x	0	0	x	1	x
1	0	1	1	1	0	x	0	1	x	x	1
1	1	0	1	1	1	x	0	x	0	1	x
1	1	1	0	0	0	x	1	x	1	x	1

(b) Excitation table

Q_3	Q_2Q_1			
	00	01	11	10
0	0	1	3	2
			1	
1	4	5	7	6
	x	x	x	x

$J_3 = Q_2Q_1$

Q_3	Q_2Q_1			
	00	01	11	10
0	0	1	3	2
	x	x	x	x
1	4	5	7	6
			1	

$K_3 = Q_2Q_1$

Q_3	Q_2Q_1			
	00	01	11	10
0	0	1	3	2
		1	x	x
1	4	5	7	6
		1	x	x

$J_2 = Q_1$

Q_3	Q_2Q_1			
	00	01	11	10
0	0	1	3	2
	x	x	1	
1	4	5	7	6
	x	x	1	

$K_2 = Q_1$

Design of Synchronous 3- Bit Up Counter



Q ₃	Q ₂ Q ₁			
	00	01	11	10
0	0	1	3	2
1	x	x	7	6

$J_3 = Q_2Q_1$

Q ₃	Q ₂ Q ₁			
	00	01	11	10
0	x	x	x	x
1			1	

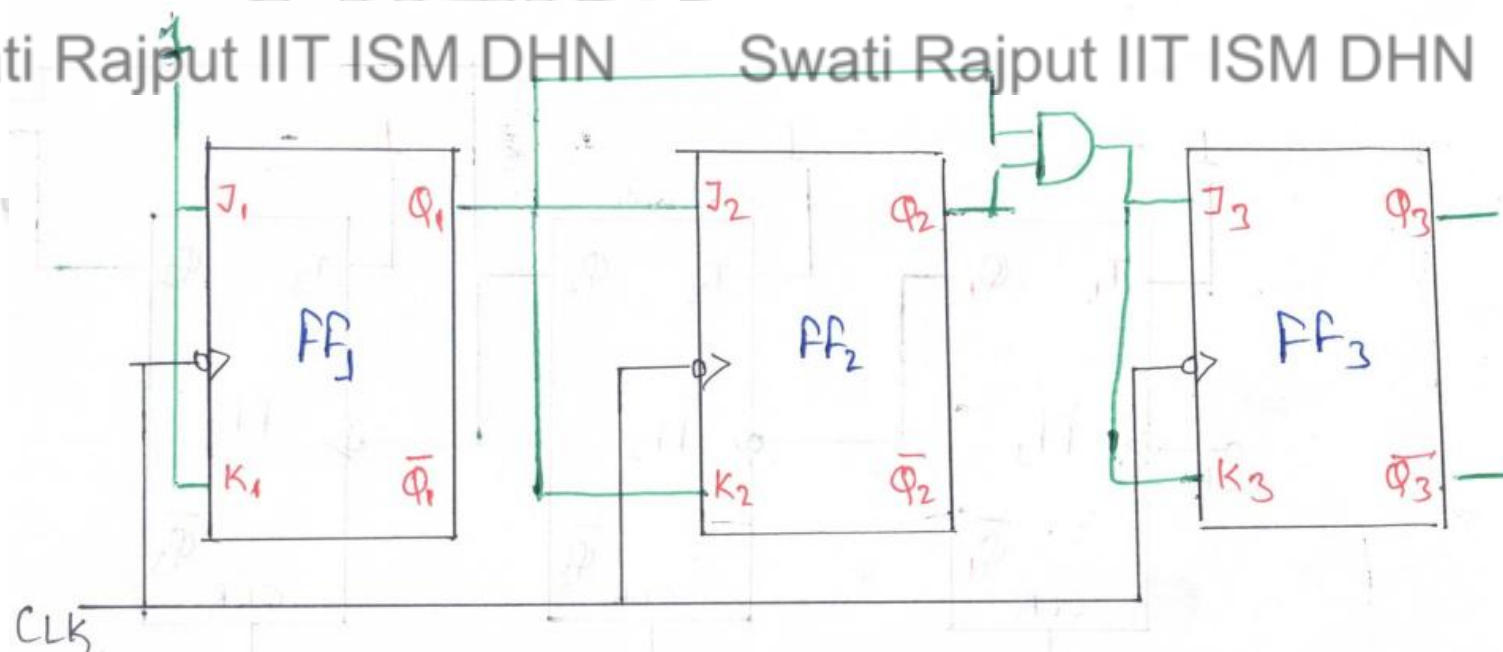
$K_3 = Q_2Q_1$

Q ₃	Q ₂ Q ₁			
	00	01	11	10
0	0	1	x	x
1	x	1	x	x

$J_2 = Q_1$

Q ₃	Q ₂ Q ₁			
	00	01	11	10
0	x	x	1	
1	x	x	1	

$K_2 = Q_1$

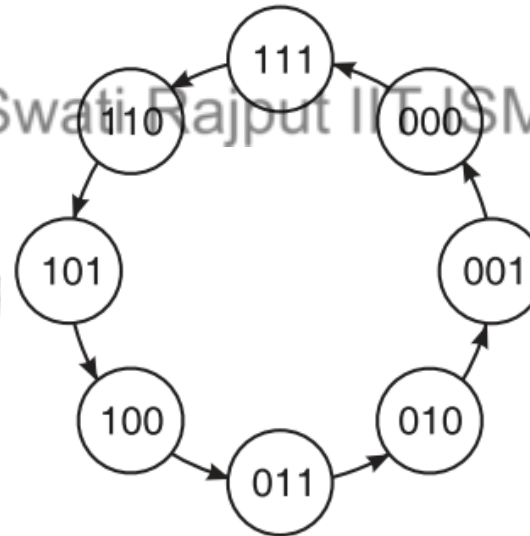


Design of Synchronous 3-Bit Down Counter



Step 1. Determine the number of flip-flops required: A 3-bit down-counter requires 3 flip-flops. The counting sequence is 000, 111, 110, 101, 100, 011, 010, 001, 000,

Step 2. Draw the state diagram: The state diagram of the 3-bit down-counter is drawn as shown below



(a) State diagram

Design of Synchronous 3-Bit Down Counter



Step 3. Select the type of flip-flops and draw the excitation table: JK flip-flops are selected and the excitation table of a 3-bit down-counter using J-K flip-flops is drawn as shown

PS			NS			Required excitations					
Q ₃	Q ₂	Q ₁	Q ₃	Q ₂	Q ₁	J ₃	K ₃	J ₂	K ₂	J ₁	K ₁
0	0	0	1	1	1	1	x	1	x	1	x
1	1	1	1	1	0	x	0	x	0	x	1
1	1	0	1	0	1	x	0	x	1	1	x
1	0	1	1	0	0	x	0	0	x	x	1
1	0	0	0	1	1	x	1	1	x	1	x
0	1	1	0	1	0	0	x	x	0	x	1
0	1	0	0	0	1	0	x	x	1	1	x
0	0	1	0	0	0	0	x	0	x	x	1

(b) Excitation table

Excitation Table of JK Flip Flop

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

PS			NS			Required excitations					
Q ₃	Q ₂	Q ₁	Q ₃	Q ₂	Q ₁	J ₃	K ₃	J ₂	K ₂	J ₁	K ₁
0	0	0	1	1	1	1	x	1	x	1	x
1	1	1	1	1	0	x	0	x	0	x	1
1	1	0	1	0	1	x	0	x	1	1	x
1	0	1	1	0	0	x	0	0	x	x	1
1	0	0	0	1	1	x	1	1	x	1	x
0	1	1	0	1	0	0	x	x	0	x	1
0	1	0	0	0	1	0	x	x	1	1	x
0	0	1	0	0	0	0	x	0	x	x	1

(b) Excitation table

Design of Synchronous 3-Bit Down Counter



Step 4. Obtain the minimal expressions: From the excitation table it is seen that, $J_1 = K_1 = 1$, because all the entries for J_1 and K_1 are either a 1 or an X. The K-maps for excitations based on the excitation table and the minimal expressions for excitations J_3 , K_3 , J_2 , and K_2 in terms of the present outputs Q_3 , Q_2 , and Q_1 obtained by minimizing the K-maps are shown

PS			NS			Required excitations					
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	1	1	1	1	x	1	x	1	x
1	1	1	1	1	0	x	0	x	0	x	1
1	1	0	1	0	1	x	0	x	1	1	x
1	0	1	1	0	0	x	0	0	x	x	1
1	0	0	0	1	1	x	1	1	x	1	x
0	1	1	0	1	0	0	x	x	0	x	1
0	1	0	0	0	1	0	x	x	1	1	x
0	0	1	0	0	0	0	x	0	x	x	1

(b) Excitation table

Q_3	Q_2Q_1			
	00	01	11	10
0	1	1	3	2
1	x	x	x	x

$J_3 = \bar{Q}_2\bar{Q}_1$

Q_3	Q_2Q_1			
	00	01	11	10
0	x	x	x	x
1	1	1	1	1

$K_3 = \bar{Q}_2\bar{Q}_1$

Q_3	Q_2Q_1			
	00	01	11	10
0	1	1	x	x
1	1	1	x	x

$J_2 = \bar{Q}_1$

Q_3	Q_2Q_1			
	00	01	11	10
0	x	x	1	1
1	x	x	1	1

$K_2 = \bar{Q}_1$

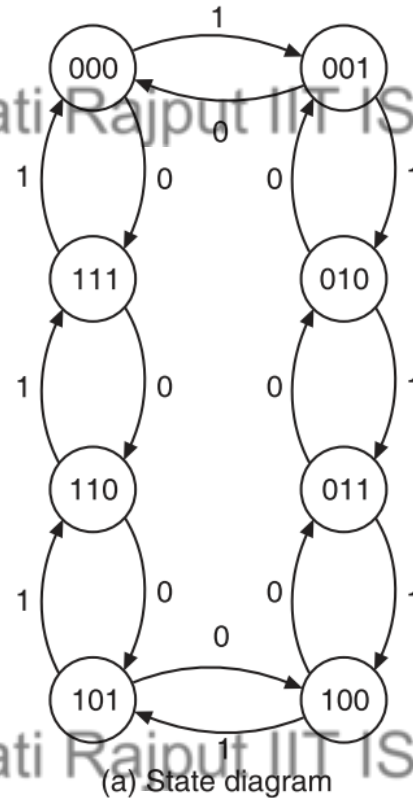
Design of Synchronous 3-Bit Up Down Counter



Step 1. Determine the number of flip-flops required: A 3-bit counter requires three FFs. It has 8 states (000, 001, 010, 011, 100, 101, 110, 111) and all the states are valid. Hence no don't cares.

For selecting up and down modes, a control or mode signal M is required. Let us say it counts up when the mode signal $M = 1$ and counts down when $M = 0$. The clock signal is applied to all the FFs simultaneously.

Step 2. Draw the state diagram: The state diagram of the 3-bit up-down counter is drawn as shown



Design of Synchronous 3-Bit Up Down Counter



Step 3. Select the type of flip-flops and draw the excitation table: JK flip-flops are selected and the excitation table of a 3-bit up-down counter using JK flip-flops is drawn as shown

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

PS			Mode	NS			Required excitations					
Q_3	Q_2	Q_1		Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	1	1	1	1	x	1	x	1	x
0	0	0	1	0	0	1	0	x	0	x	1	x
0	0	1	0	0	0	0	0	x	0	x	x	1
0	0	1	1	0	1	0	0	x	1	x	x	1
0	1	0	0	0	0	1	0	x	x	1	x	x
0	1	0	1	0	1	1	0	x	x	0	1	x
0	1	1	0	0	1	0	0	x	x	0	x	1
0	1	1	1	1	0	0	1	x	x	1	x	1
1	0	0	0	0	1	1	x	1	1	x	1	x
1	0	0	1	1	0	1	x	0	0	x	1	x
1	0	1	0	1	0	0	x	0	0	x	x	1
1	0	1	1	1	1	0	x	0	1	x	x	1
1	1	0	0	1	0	1	x	0	x	1	1	x
1	1	0	1	1	1	1	x	0	x	0	1	x
1	1	1	0	1	1	0	x	0	x	0	x	1
1	1	1	1	0	0	0	x	1	x	1	x	1

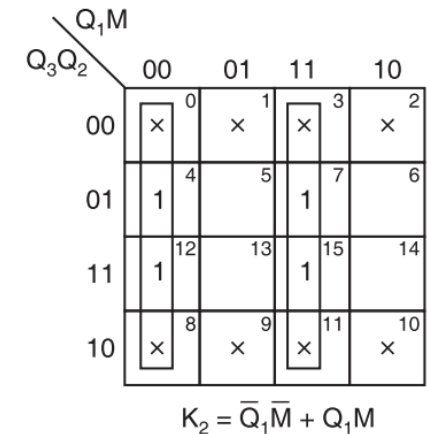
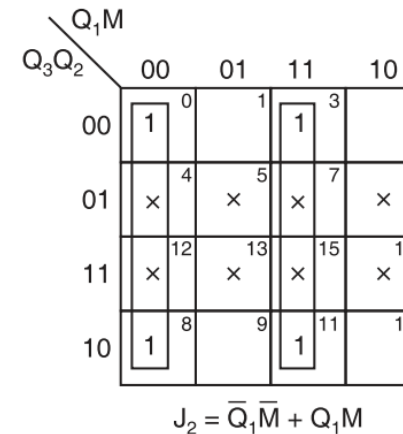
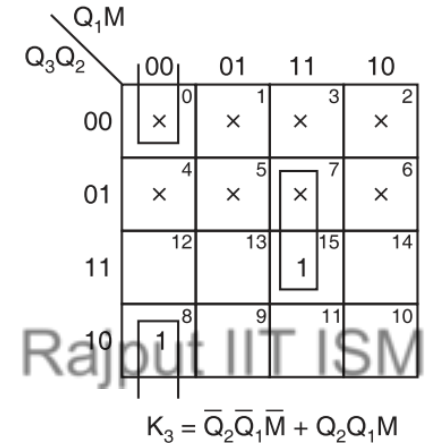
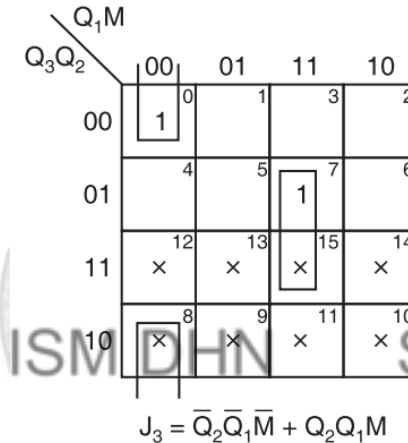
(b) Excitation table

Design of Synchronous 3-Bit Up Down Counter



Step 4. Obtain the minimal expressions: From the excitation table we can conclude that $J_1 = 1$ and $K_1 = 1$, because all the entries for J_1 and K_1 are either X or 1. The K-maps for J_3 , K_3 , J_2 and K_2 based on the excitation table and the minimal expressions obtained from them are shown

PS			Mode	NS			Required excitations					
Q_3	Q_2	Q_1	M	Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	1	1	1	1	x	1	x	1	x
0	0	0	1	0	0	1	0	x	0	x	1	x
0	0	1	0	0	0	0	0	x	0	x	x	1
0	0	1	1	0	1	0	0	x	1	x	x	1
0	1	0	0	0	0	1	0	x	x	1	1	x
0	1	0	1	0	1	1	0	x	x	0	1	x
0	1	1	0	0	1	0	0	x	x	0	x	1
0	1	1	1	1	0	0	1	x	x	1	x	1
1	0	0	0	0	1	1	x	1	1	x	1	x
1	0	0	1	1	0	1	x	0	0	x	1	x
1	0	1	0	1	1	0	x	0	1	x	x	1
1	1	0	0	1	0	1	x	0	x	1	1	x
1	1	0	1	1	1	1	x	0	x	0	1	x
1	1	1	0	1	1	0	x	0	x	0	x	1
1	1	1	1	0	0	0	x	1	x	1	x	1

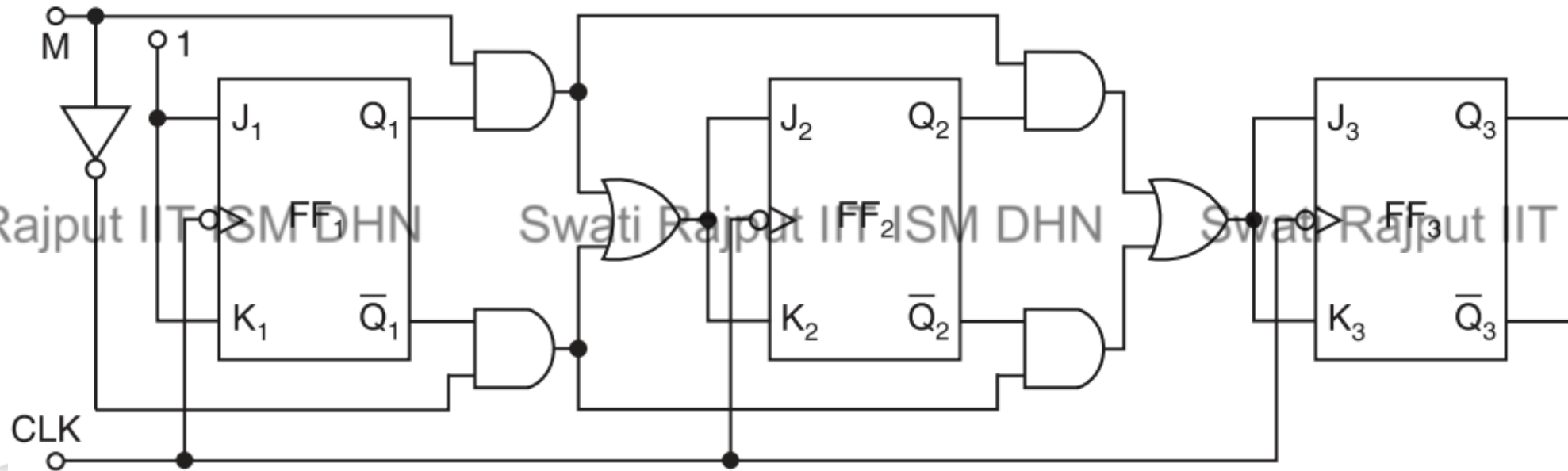


(b) Excitation table

Design of Synchronous 3-Bit Up Down Counter



Step 5. Draw the logic diagram: A logic diagram using those minimal expressions can be drawn as shown

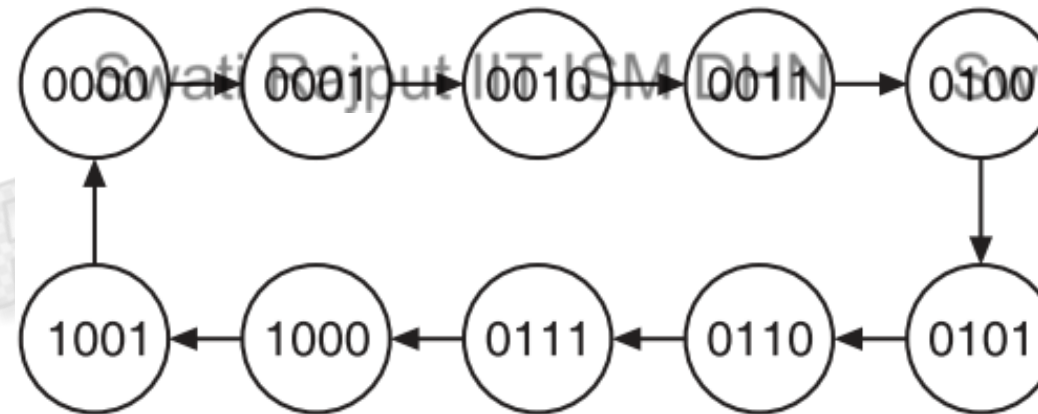


Design of Synchronous BCD Counter



Step 1. Determine the number of flip-flops required: A BCD counter is nothing but a mod-10 counter. It is a decade counter. It has 10 states (0000 through 1001). It requires $n = 4$ FFs ($N \leq 2^n$, i.e. $10 \leq 2^4$). Four FFs can have 16 states. After the tenth clock pulse, the counter resets. So, states 1010 through, 1111 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. Draw the state diagram: The state diagram of the BCD counter is drawn as shown



(a) State diagram

Design of Synchronous BCD Counter



Step 3. The type of flip-flops and the excitation table: JK flip-flops are selected and the excitation table of the BCD counter using J-K FFs is drawn as shown

Excitation Table of JK Flip Flop

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

PS				NS				Required excitations							
Q_4	Q_3	Q_2	Q_1	Q_4	Q_3	Q_2	Q_1	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	0	1	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1

(b) Excitation table

Design of Synchronous BCD Counter



Step 4. The minimal expressions: The K-maps for the excitations of FFs $J_4, K_4, J_3, K_3, J_2, K_2, J_1$ and K_1 in terms of the outputs of the FFs Q_4, Q_3, Q_2 and Q_1 , based on the excitation table, their minimization and the minimal expressions obtained from them are shown

PS				NS				Required excitations							
Q_4	Q_3	Q_2	Q_1	Q_4	Q_3	Q_2	Q_1	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1

(b) Excitation table

K-map for J_4 :

Q_2Q_1	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Minimal expression: $J_4 = Q_3Q_2Q_1$

K-map for K_4 :

Q_2Q_1	00	01	11	10
00	x	x	x	x
01	x	x	x	x
11	x	x	x	x
10	x	1	x	x

Minimal expression: $K_4 = Q_1$

K-map for J_3 :

Q_2Q_1	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Minimal expression: $J_3 = Q_2Q_1$

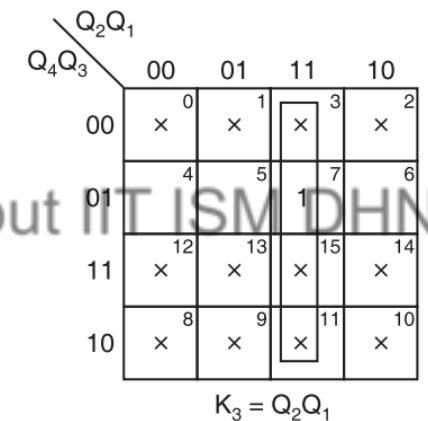
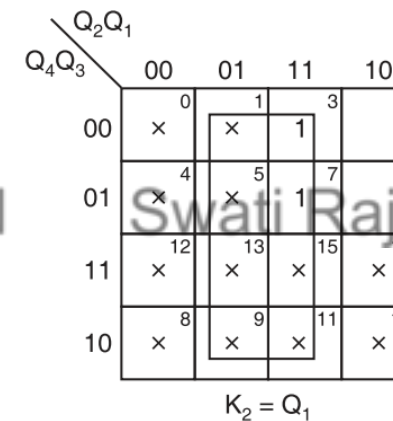
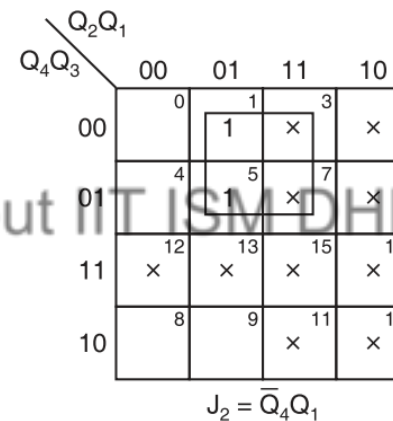
Design of Synchronous BCD Counter



Step 4. The minimal expressions: The K-maps for the excitations of FFs $J_4, K_4, J_3, K_3, J_2, K_2, J_1$ and K_1 in terms of the outputs of the FFs Q_4, Q_3, Q_2 and Q_1 , based on the excitation table, their minimization and the minimal expressions obtained from them are shown

PS				NS				Required excitations							
Q_4	Q_3	Q_2	Q_1	Q_4	Q_3	Q_2	Q_1	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1

(b) Excitation table



Swati Rajput IIT ISM DHN Swati Rajput IIT ISM DHN Swati Rajput IIT ISM DHN

Design of Synchronous BCD Counter



- ❖ The state diagram and the table to check for lock-out are shown in Figure.
- ❖ The table to check for lock-out shows that, if the counter finds itself in an invalid state initially, it moves to a valid state after one or two clock pulses and then counts in the normal way. Therefore, the counter is self-starting.

$$J_4 = Q_3 Q_2 Q_1$$

$$K_4 = Q_1$$

$$J_3 = Q_2 Q_1$$

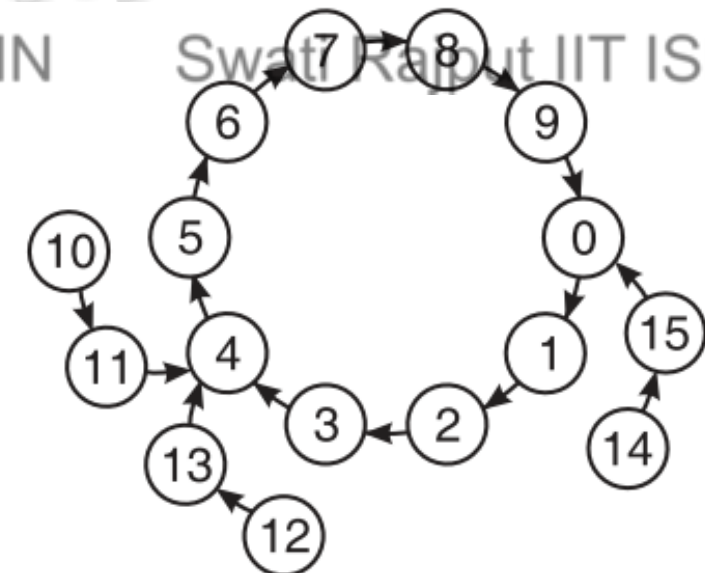
$$J_2 = \bar{Q}_4 Q_1$$

$$K_2 = Q_1$$

$$K_3 = Q_2 Q_1$$

PS				Present inputs								NS			
Q_4	Q_3	Q_2	Q_1	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1	Q_4	Q_3	Q_2	Q_1
1	0	1	0	0	0	0	0	0	0	1	1	1	0	1	1
1	0	1	1	0	1	1	1	0	1	1	1	0	1	0	0
1	1	0	0	0	0	0	0	0	0	1	1	1	1	0	1
1	1	0	1	0	1	0	0	0	1	1	1	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	1	1	1	1	1
1	1	1	1	1	1	1	1	0	1	1	1	0	0	0	0

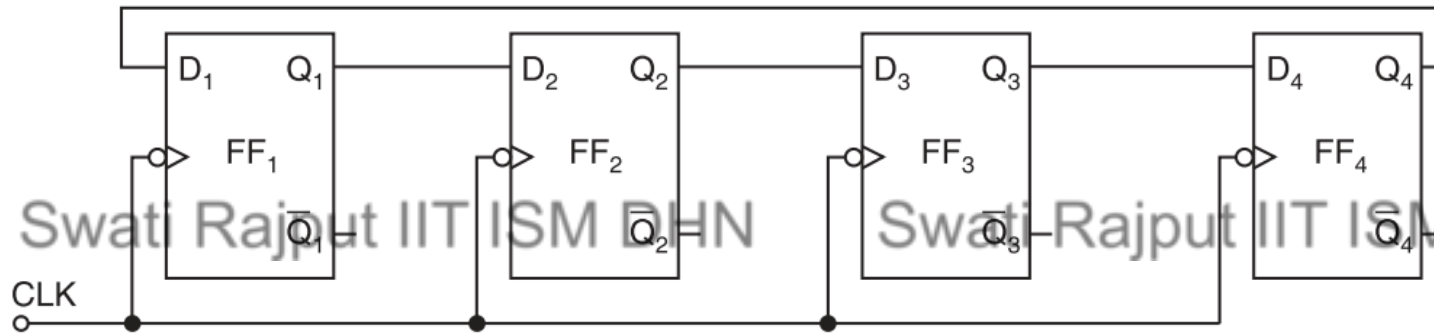
(a) Table to check for lock-out



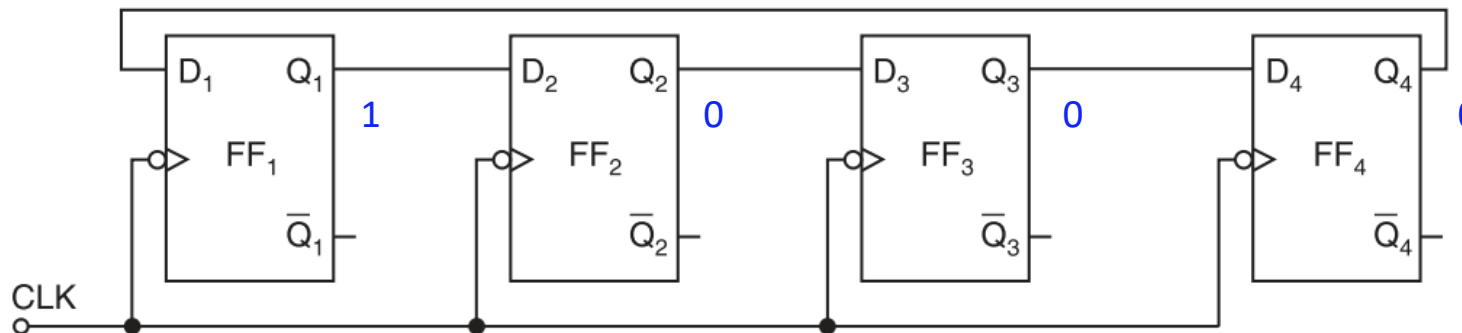
(b) State diagram

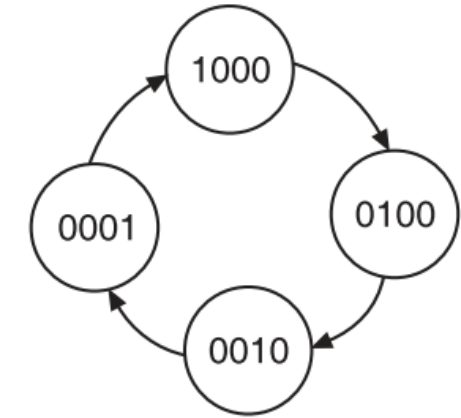
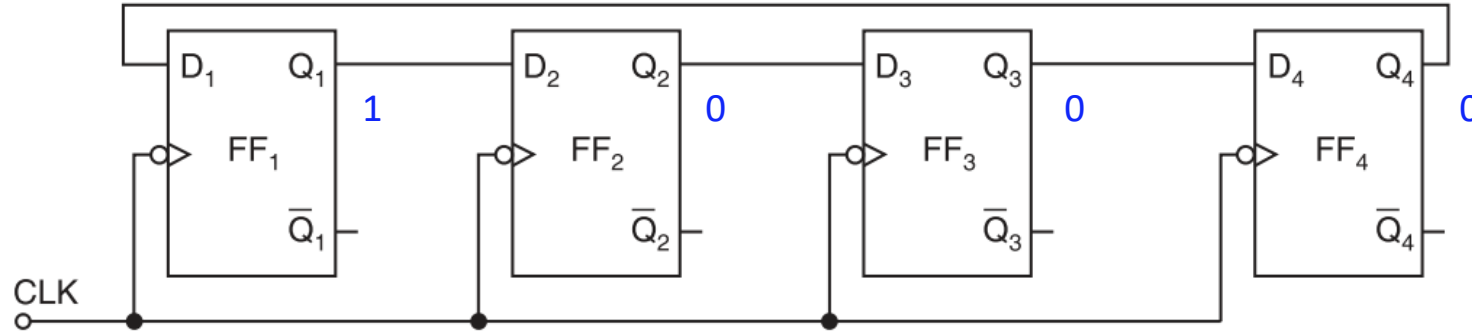


- ❖ This is the simplest shift register counter.
- ❖ The basic ring counter using D FFs is shown in Figure.
- ❖ The FFs are arranged as in a normal shift register, i.e. the Q output of each stage is connected to the D input of the next stage, but the Q output of the last FF is connected back to the D input of the first FF such that the array of FFs is arranged in a ring and, therefore, the name *ring counter*.

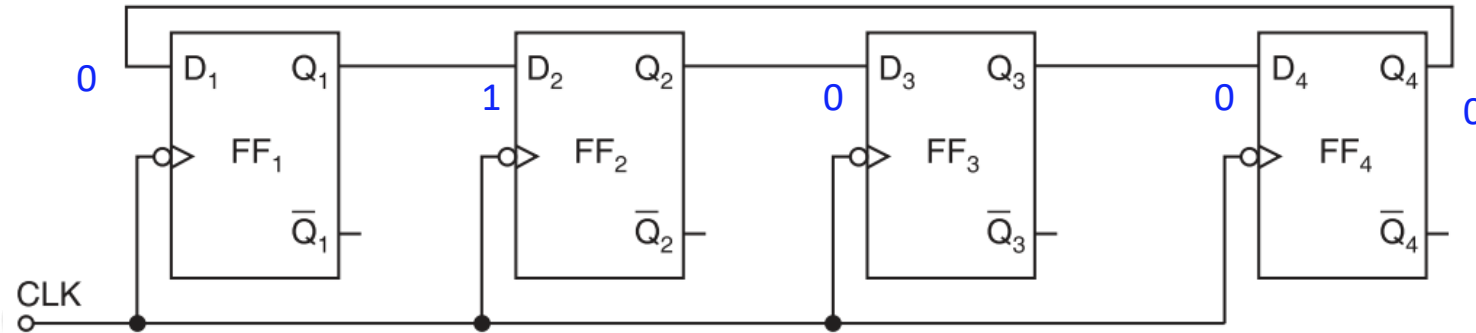


- ❖ Initially, the first FF is preset to a 1. So, the initial state is 1000, i.e. $Q_1 = 1$, $Q_2 = 0$, $Q_3 = 0$ and $Q_4 = 0$.





(a) State diagram

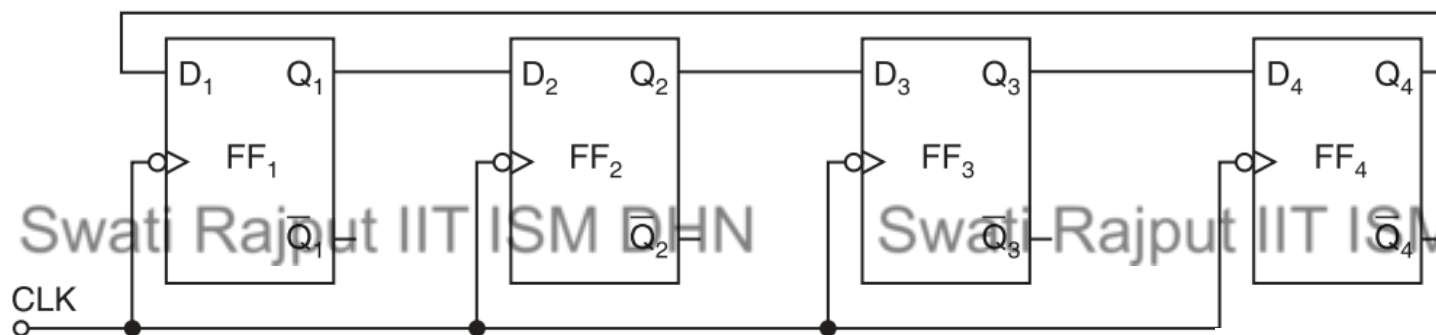


Q ₁	Q ₂	Q ₃	Q ₄	After clock pulse
1	0	0	0	0
0	1	0	0	1
0	0	1	0	2
0	0	0	1	3
1	0	0	0	4
0	1	0	0	5
0	0	1	0	6
0	0	0	1	7

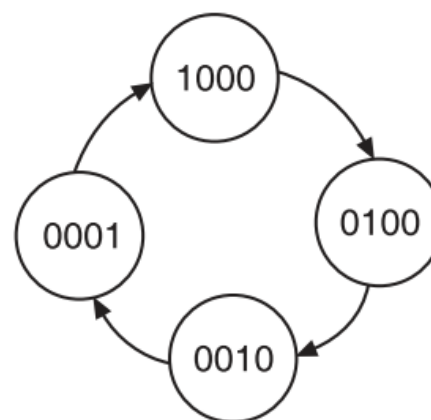
(b) Sequence table



- ❖ This is the simplest shift register counter.
- ❖ The basic ring counter using D FFs is shown in Figure.
- ❖ The FFs are arranged as in a normal shift register, i.e. the Q output of each stage is connected to the D input of the next stage, but the Q output of the last FF is connected back to the D input of the first FF such that the array of FFs is arranged in a ring and, therefore, the name *ring counter*.



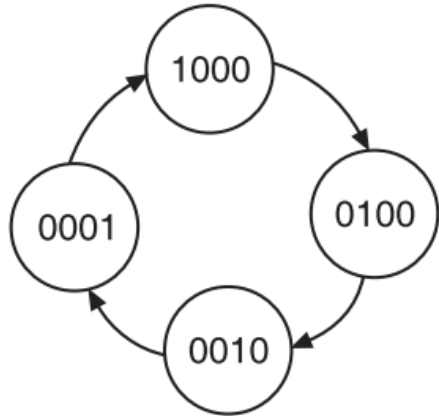
- ❖ Initially, the first FF is preset to a 1. So, the initial state is 1000, i.e. $Q_1 = 1$, $Q_2 = 0$, $Q_3 = 0$ and $Q_4 = 0$.
- ❖ After each clock pulse, the contents of the register are shifted to the right by one bit and Q_4 is shifted back to Q_1 .
- ❖ The sequence repeats after four clock pulses.
- ❖ The number of distinct states in the ring counter, i.e. the mod of the ring counter is equal to the number of FFs used in the counter.
- ❖ An n -bit ring counter can count only n bits, whereas an n -bit ripple counter can count 2^n bits



(a) State diagram

Q_1	Q_2	Q_3	Q_4	After clock pulse
1	0	0	0	0
0	1	0	0	1
0	0	1	0	2
0	0	0	1	3
1	0	0	0	4
0	1	0	0	5
0	0	1	0	6
0	0	0	1	7

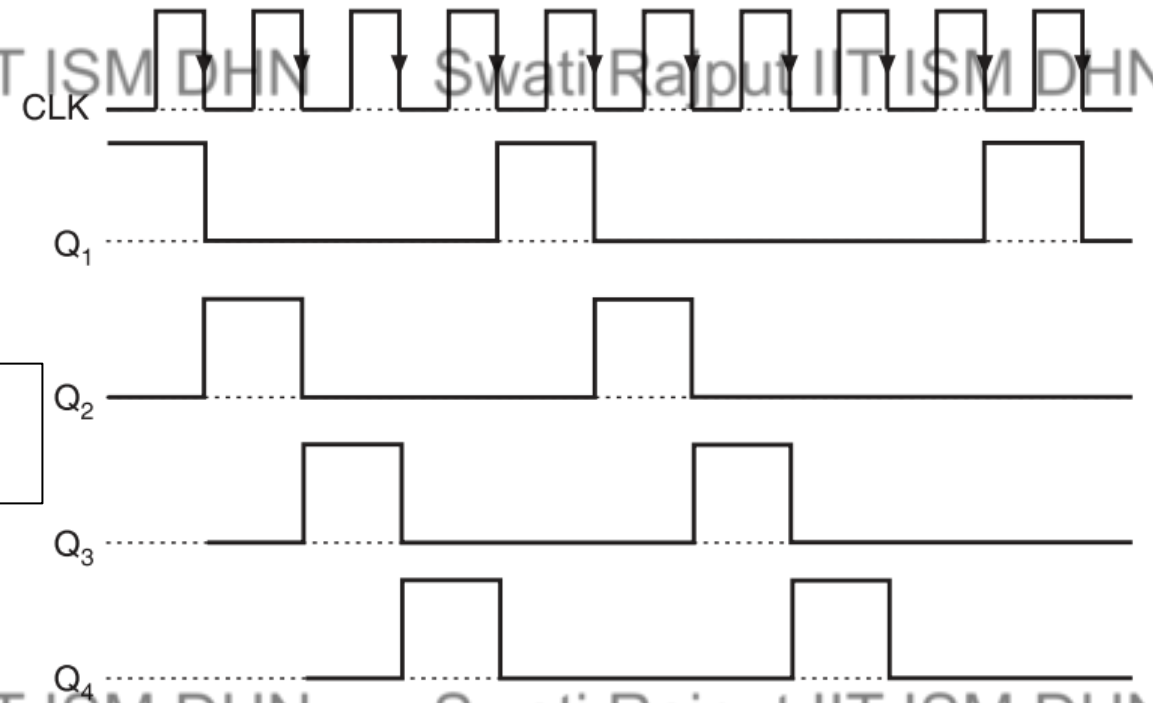
(b) Sequence table



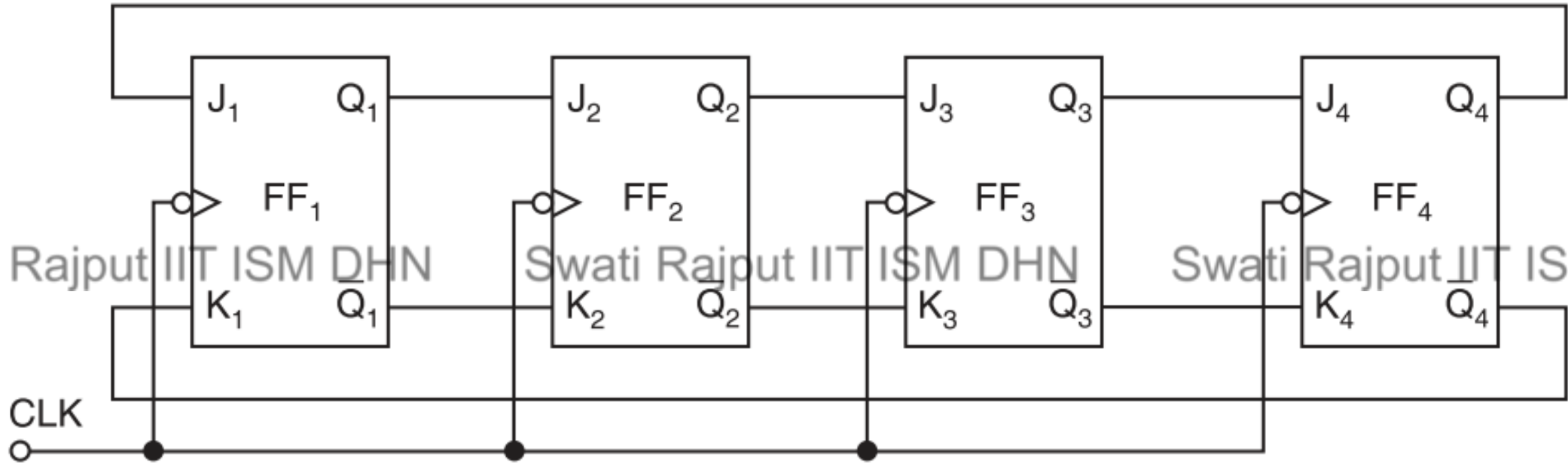
(a) State diagram

Q_1	Q_2	Q_3	Q_4	After clock pulse
1	0	0	0	0
0	1	0	0	1
0	0	1	0	2
0	0	0	1	3
1	0	0	0	4
0	1	0	0	5
0	0	1	0	6
0	0	0	1	7

(b) Sequence table



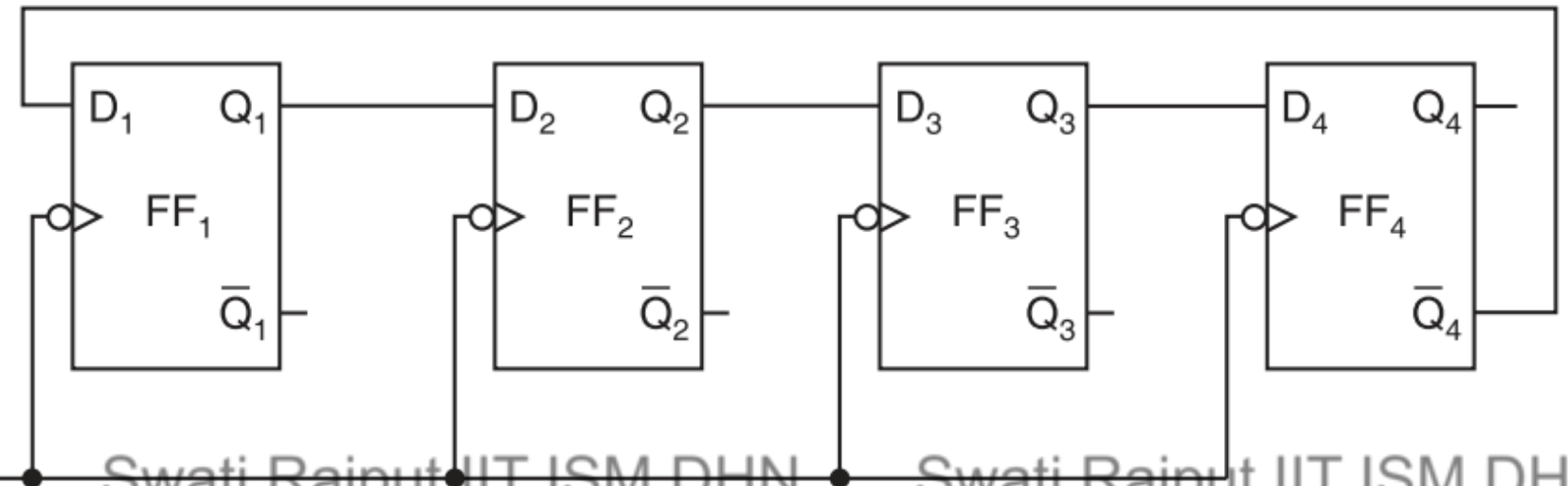
Since it is entirely a synchronous operation and requires no gates external to FFs, it has the further advantage of being very fast.



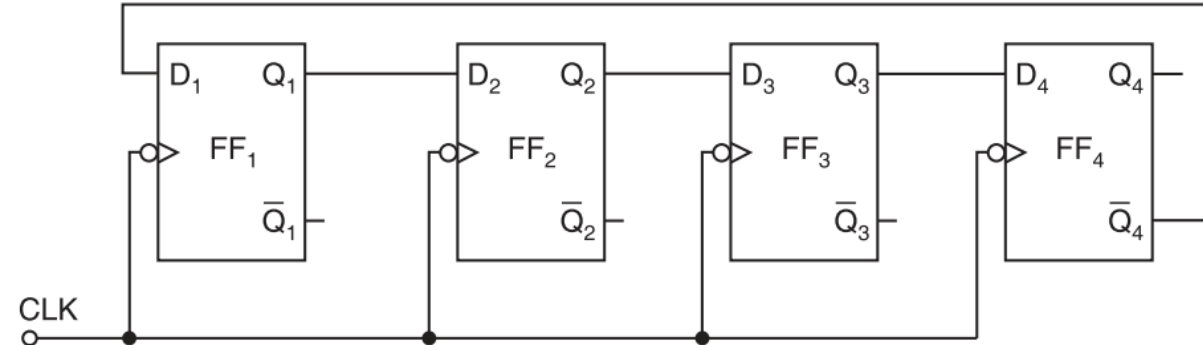
Twisted Ring Counter (Johnson Counter)



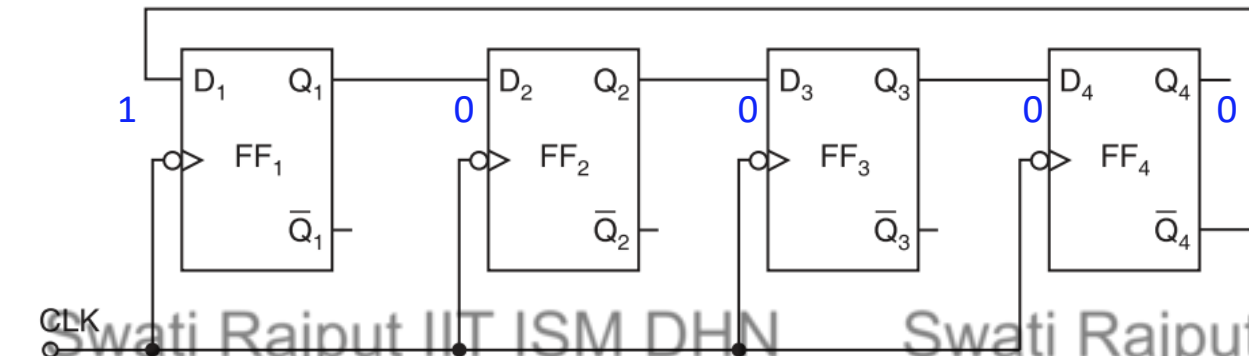
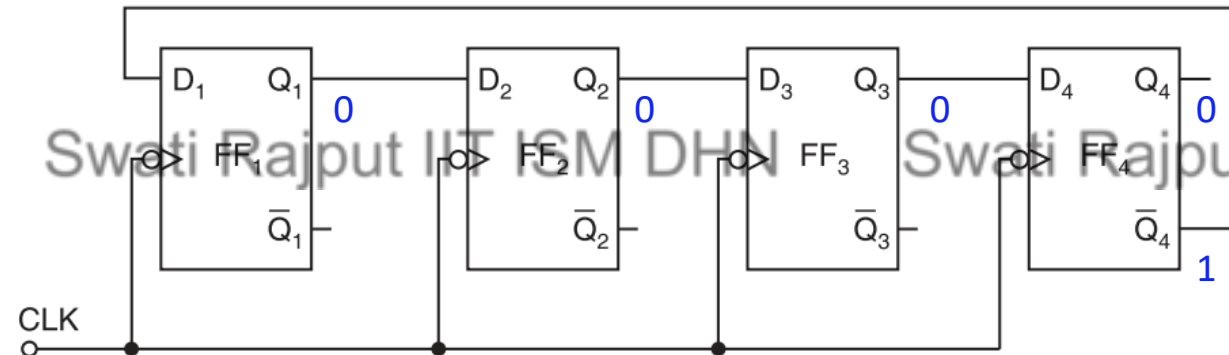
- ❖ This counter is obtained from a serial-in, serial-out shift register by providing feedback from the inverted output of the last FF to the D input of the first FF.
- ❖ The Q output of each stage is connected to the D input of the next stage, but the $\bar{Q}$ output of the last stage is connected to the D input of first stage, therefore, the name *twisted ring counter*. This feedback arrangement produces a unique sequence of states.



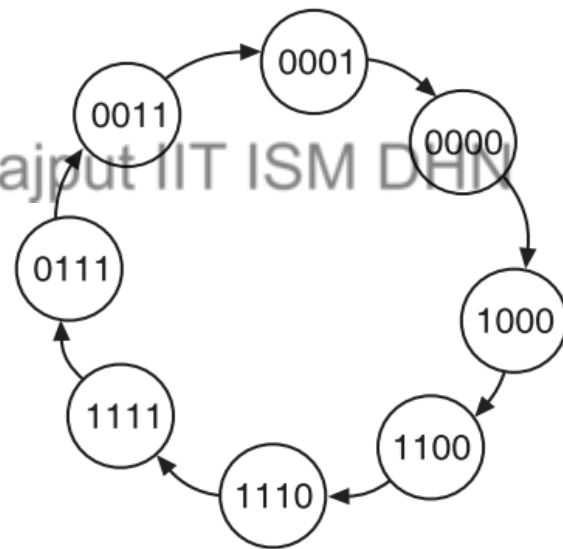
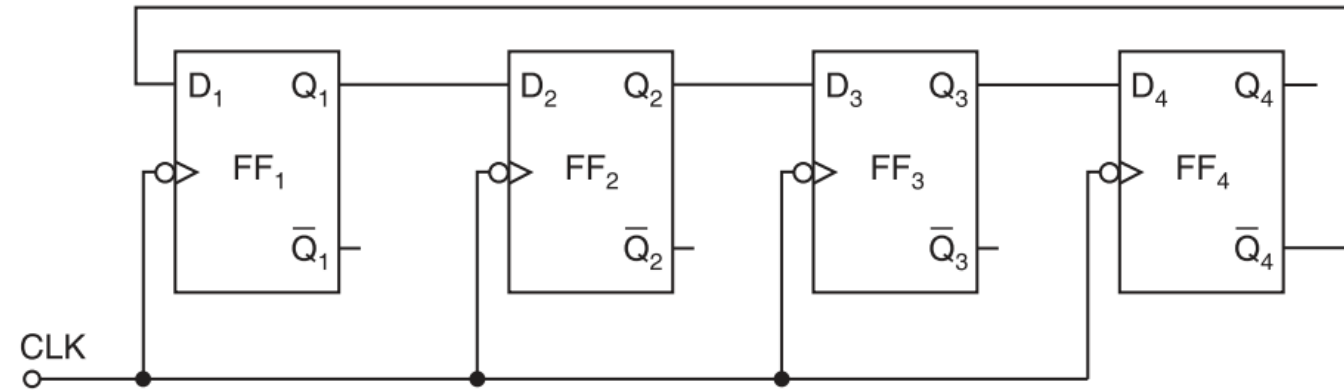
Twisted Ring Counter (Johnson Counter)



- ❖ Let initially all the FFs be reset, i.e. the state of the counter be 0000.
- ❖ After each clock pulse, the level of Q_1 is shifted to Q_2 , the level of Q_2 to Q_3 , Q_3 to Q_4 and the level of Q_4 to Q_1 . This sequence is repeated after every eight clock pulses.



Twisted Ring Counter (Johnson Counter)



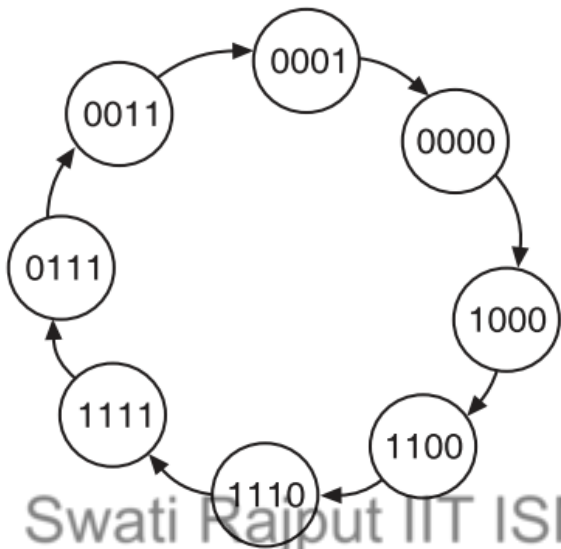
(a) State diagram

Q ₁	Q ₂	Q ₃	Q ₄	After clock pulse
0	0	0	0	0
1	0	0	0	1
1	1	0	0	2
1	1	1	0	3
1	1	1	1	4
0	1	1	1	5
0	0	1	1	6
0	0	0	1	7
0	0	0	0	8
1	0	0	0	9

(b) Sequence table

- ❖ Let initially all the FFs be reset, i.e. the state of the counter be 0000.
- ❖ After each clock pulse, the level of Q₁ is shifted to Q₂, the level of Q₂ to Q₃, Q₃ to Q₄ and the level of Q₄ to Q₁. This sequence is repeated after every eight clock pulses.

Twisted Ring Counter (Johnson Counter)



(a) State diagram

Q_1	Q_2	Q_3	Q_4	After clock pulse
0	0	0	0	0
1	0	0	0	1
1	1	0	0	2
1	1	1	0	3
1	1	1	1	4
0	1	1	1	5
0	0	1	1	6
0	0	0	1	7
0	0	0	0	8
1	0	0	0	9

(b) Sequence table

- ❖ An n FF Johnson counter can have $2n$ unique states and can count up to $2n$ pulses. So, it is a mod- $2n$ counter. It is more economical than the normal ring counter.

